P. Krishna Chaitanya

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PROFILE

Undergraduate physics student (Honours with Research) with hands-on experience in solid-state synthesis of lead-free piezoelectric ceramics, solid rocket motor hardware, and first steps in aircraft modelling for flight stability. Pursuing a Flight Mechanics minor alongside the physics degree.

EDUCATION

Sri Sathya Sai Institute of Higher Learning (SSSIHL) — Prasanthi Nilayam, India

B.S. (Hons. with Research) Physics | 2025 – present

- CGPA: 7.2 / 10 (cumulative, Semesters 1–2); Semester 3 in progress
- Minor: Flight Mechanics
- Coursework: Mathematical Physics I, Electromagnetism, Computational Techniques in Physics; Computational Physics, Electronics and Mechanics labs

RESEARCH EXPERIENCE

Research Intern — School of Engineering Sciences & Technology (SEST), University of Hyderabad

Under Prof. Dibakar Das | 18 May – 18 June 2026 | Hyderabad, India

- Synthesised lead-free $K_{0.48}Na_{0.52}NbO_3$ (KNN) piezoceramics by the solid-state route; did the weighing, wet ball milling, calcination, pressing, binder burnout and sintering myself, and prepared the XRD samples.
- Compared two calcination batches (825 °C / 4 h and 900 °C / 2 h); XRD showed the main orthorhombic KNN perovskite phase with a minor $K_2Nb_4O_{11}$ secondary phase.
- Sintering at 1000 °C / 2 h gave no densification (63–64%); at 1100 °C / 4 h the best pellets reached 88.7% and 91.4% geometric density.

PROJECTS

KNSU Solid Rocket Motor — Design, Fabrication and Test (team project, ongoing)

- Designing and building a sugar-propellant (KNSU) motor with an aluminium casing, flown in a PVC airframe with through-the-wall fins.
- First launch attempt did not lift off; diagnosed insufficient thrust as the cause.
- Redesigned the nozzle for a scaled 30 mm variant (70:30 plaster-of-Paris/cement composite with a metal throat insert), with STL models and a written design basis.
- Planned next: an instrumented static fire to measure the thrust curve, and a CP/CG stability check before the next flight.

Aerodynamic Modelling in OpenVSP (Aug 2026)

- Built an OpenVSP aircraft model and ran a first VSPAERO span-load analysis at α = 0°, 5° and 10°; a clean rerun is planned.
- Completed all 30 lectures of the NPTEL course *Aircraft Stability and Control* (Prof. A. K. Ghosh, IIT Kanpur); certification examination on 25 October 2026.

Computational Physics in Python

- Lab assignments in interpolation, cubic splines, a 2D spline surface fit, linear systems and eigenvalue problems, plus plotting and analysis of my own lab data.

AI-DIRECTED SOFTWARE BUILDS

- Directed AI coding agents (Claude) through builds where I set the scope and design and the AI wrote the code: a Unity 6 game prologue taken through four versions, a Windows voice assistant with local speech-to-text and OCR, HandMouse 2, a webcam hand-control app for Windows (MediaPipe), a bio and opener generator, a CLI habit tracker, and a pipeline that turns my lab data into report PDFs.

ACADEMIC PRESENTATION

Poster — Red-light interaction with mitochondria for glucose regulation | NUPC, IISER Tirupati | with Amar Manas Akula

- A simplified mitochondrial signalling model (MATLAB SimBiology) and a tissue optical fluence simulation of red-light photobiomodulation, compared with clinical data showing lower blood glucose after 670 nm light.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib): numerical methods and lab-data analysis.
- **Scientific and engineering tools:** OpenVSP / VSPAERO, MATLAB SimBiology, Scilab, Excel / Google Sheets, AI coding agents (Claude Code).
- **Experimental:** solid-state synthesis, ball milling, pellet pressing and sintering, geometric density measurement, XRD sample preparation, analog and digital electronics, error analysis.
- **Engineering:** aerodynamics and flight stability (coursework), propulsion fundamentals, fabrication and prototyping.
- **Communication:** literature review, technical writing, poster presentation.
- **Languages:** English — fluent (medium of instruction); German — A2 in progress, working towards B1.